Network and graph analytics

Scoring rings, not single payments

Status: Draft v0.1 (2026-11-04), for Steve's review. Extension 2 of 10 (see the extensions overview). Reference code: `analytics/forensic/network.py` (`communities`, `cluster_features`, `cluster_score`, `rank_clusters`), tested on synthetic data only.

Bottom line

Organized fraud runs through groups of entities that look ordinary one at a time: twenty hospices that each bill a plausible amount, eight PPP borrowers with plausible payrolls, five "small businesses" that each win a few set-asides. What gives them away is what they share: an address, an officer, a phone line, a bank account, a preparer. The forensic methods handbook already builds the shared-attribute graph and propagates risk from known bad actors. This extension adds the step that turns a graph into a work queue: **split the graph into communities, describe each community with a few features, and rank communities by a calibrated score**, so investigators open the ring, not one member of it.

1. Graph

Nodes are entities (businesses, providers, borrowers). An edge joins two entities that share an attribute: address, officer or owner, phone, email, bank account, registered agent, or IP or device in Phase 2. Edge weight follows the handbook (section 8): each shared attribute adds 1 / log(1 + degree), so a phone shared by two firms counts far more than a registered-agent address shared by 4,000. Attributes on the benign-hub list (registered agents, mail centers, hospitals, large payroll firms) and attributes with more than 25 entities are dropped.

2. From components to communities

Connected components chain together through single weak links, so one shared accountant can glue two unrelated rings into one blob. Weighted label propagation splits a component into groups that are densely tied inside and loosely tied outside. It is deterministic for a given seed, needs no tuning and runs in near-linear time, which matters at SAM.gov scale (several hundred thousand active registrations). Louvain or Leiden are drop-in alternatives in a graph database.

3. Cluster features

Feature	Definition	Why it matters
Size	Entities in the community	Rings are usually 3 to 50 entities
Density	Internal edges / possible edges	Rings are tightly interconnected; franchises are star-shaped
Bank share	Share of members sharing a bank account with another member	The strongest single sign of common control
New share	Share first seen in the last two years	Rings form fast around a new program or rule change
Dollars	Total paid to the community	Sets priority among similar scores
Seed risk	Highest risk-propagation score from excluded, debarred or charged entities	Known bad actors nearby

4. Score

A logistic score combines the features, with size and dollars on log scales so a 200-outlet franchise does not outrank a six-firm ring that shares bank accounts. The default weights in `CLUSTER_WEIGHTS` are starting values. Before any cluster is referred they are refit on confirmed rings (DOJ case releases, OIG reports, the SBA case set) using the likelihood ratio method in the handbook section 12, and the score must meet the review plan's release bar: known rings at least five times more common in the top 5% than in the population.

5. What an investigator sees

For each ranked cluster the tool shows: members and their awards or payments; the shared attributes that tie them, strongest first; a small graph drawing; dates of formation; any seed (excluded or charged entity) within two hops; and the reason codes behind the score. The tool preview pages on the site follow this layout.

6. Where it applies first

Roadmap idea	Graph	Seeds
HHS-1, HHS-2 hospice and home health	Owners, managing employees, addresses from CMS enrollment files	LEIE, revoked providers, DOJ defendants
SBA-1, SBA-3 loan rings	Borrower addresses, lenders, agents, NAICS	Charged PPP and EIDL defendants
DOD-1, VA-1 set-aside fronts	SAM.gov officers, addresses, parent firms	SAM exclusions, SBA protests sustained
TREAS-2 ERC and EITC promoters	Preparers and promoters in charged cases	DOJ Tax Division cases
X-4 hidden common control	All of the above, linked through cross-agency matching	All exclusion lists

7. Limits



- Public data shows registrations, not control. A shared officer may be a part-time bookkeeper; the score asks a question, it does not answer one.
- Rings adapt by spreading attributes thin (one phone per firm). The red-team playbook sets out how the features respond.
- Bank accounts are the strongest feature and are not public; until Phase 2 the bank-share feature is zero and the score leans on officers, phones and addresses.

8. Next steps

1. Fit weights on the SBA confirmed set (weeks 5 to 6).
2. Run on HHS hospice enrollment data alongside HHS-1 (weeks 6 to 8).
3. Add a cluster view to the HHS-1 tool preview.

